

Supplementary Evaluation Results for VERIEDGE: Verifiability-Constrained Placement for Heterogeneous Edge LLM Inference

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This supplement provides evaluation details that support the main paper without extending its claims. It reports the measured verifier-profile matrix, placement replay inputs, selective-delivery scaling, sketch robustness checks, known sketch blind spots, overhead breakdowns, and full policy-composition views.

1 Measured Verifier Profiles Used by Placement Replay

Table 1 reports the verifier-profile matrix used by the closed-loop placement replay. Each row is a profile key: stack pair, sketch mode, and checked C1–C3 boundaries. The risk class is the descriptive class used by the risk-composition plots; task admission still applies the explicit α , β , $S^{\max}$, $C^{\max}$ feasibility test. Scale TPR is reported as a diagnostic for the cosine-sketch blind spot, while admission uses held-out honest FPR and primary material-tamper TPR.

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2 Placement Replay Inputs and Policy Semantics

The replay is a controlled decision replay rather than a live marketplace trace. It fixes three inputs: the verifier-profile matrix in Table 1, a candidate-placement table, and a workload table. The candidate table contains 13 placements: one homogeneous F/F control and 12 heterogeneous LAN/WAN variants over A/B, A/B-RTXint8, A/C, A/D, B/D, and E/F. Each candidate records a provider group, a C1–C3 shard map, a stack-pair label, network and execution latency estimates, availability cost, reputation, capacity feasibility, and backend-mix class. The workload table contains 400 records: 200 evaluation prompts replayed as a low-pressure single-arrival workload and the same 200 prompts replayed as a queued-8 burst workload. Table 2 summarizes these fixed replay inputs.

Table 3 reports the queued-8 replay numbers summarized in the main paper. The main text uses the figure and key deltas for the placement narrative; this table preserves exact policy metrics for comparison.

For each task and policy, the replay first filters candidates by capacity and group-size bounds. A candidate and sketch choice then looks up its measured profile row. The selected choice is admissible when $FPR \leq \alpha$, primary-material $TPR \geq \beta$, trace bytes are within $S^{\max}$, and challenge-time verifier latency is within $C^{\max}$. If a policy selects a candidate whose profile violates any of these checks, the admitted task contributes to infeasible usage. False-dispute risk is computed as $0.10 \times FPR$ for the selected profile. Expected challenge cost is computed analogously as 0.10 times the measured honest verifier latency.

The policy families differ in when they use these profiles. Network-aware baselines first choose the lowest-latency candidate from network and execution estimates, then attach a fixed sketch. Risk-constrained policies apply the hard verifiability admission test before ranking. Adaptive-verifier policies choose the smallest feasible sketch for a selected candidate, or jointly choose a candidate and sketch under the same feasibility test. Queue-aware policies add current candidate wait time, $\max(0, \text{busyUntil} - \text{arrival})$, to the placement score before choosing a candidate. Replay latency is modeled service time plus queue wait, and goodput is the number of admitted tasks divided by final completion time.

3 Risk Composition Behind Placement Decisions

Figure 1 decomposes which verifier-profile risk classes are actually selected by each placement policy. This view complements the main paper’s placement-frontier plots: latency-oriented network and queue-aware policies can improve modeled service time or queueing behavior while still selecting profile keys that violate the task’s FPR/TPR target. Risk-constrained and adaptive-verifier policies remove those choices before task disclosure, either by filtering the placement or by upgrading the sketch mode. The same pattern appears in both arrival regimes. At $\alpha = 0.10$, network-aware projcos4 has infeasible usage 1.000 in both the single and queued-8 replay, while risk-constrained and adaptive variants keep infeasible usage at 0.000. In the queue-aware replay, queue-aware network placement reaches the highest goodput, but still uses infeasible profile keys for 93.5% of admitted tasks; queue-aware adaptive placement keeps infeasible usage at zero while preserving nearly the same goodput.

4 Selective Delivery and Placement-Defined Disclosure

The main paper reports the central 100MB selective-delivery readout. Here we include the exact modeled LAN/WAN medians in Table 4 and corresponding curves in Figure 2. The experiment compares replicated payload delivery (RPD), which sends one full encrypted payload per selected provider, against privacy-preserving delivery (PPD), which publishes one encrypted object and releases small locator-and-key packages after the placement commitment. Both modes encrypt the payload; the difference is whether requester-side payload transfer scales with selected group size.

Live local-store validation. The LAN/WAN sweep is a calibrated delivery-path model, so we also ran a local store supplement to validate the disclosure-path assumptions. Each of the 240 local-store runs created a fresh ciphertext, derived the object key from the actual SHA-256 content hash, had providers fetch the object, and hash-checked the fetched ciphertext. This supplement is not used as the LAN/WAN latency source; it rules out cache reuse and confirms that the measured PPD path uses fresh stored objects and targeted access material. Table 5 reports the exact local-store medians.

5 Projected-Scalar Bridge Stability

Figure 4 checks whether the projscalar1_abs bridge in the main evaluation is an artifact of one random seed. Across A/B, A/B-RTXint8, A/C, A/D, B/D, and E/F with five seeds, every selected operating point remains admissible under the $\alpha = 0.10, \beta = 0.90$ profile, with mean held-out FPR 0.0073, max FPR 0.025, and minimum primary-material TPR 1.000. This supports using the sketch as a norm-sensitive companion to projected-token cosine TSTC. It is a diagnostic

and hybrid-verifier component rather than a replacement for projected-token TSTC.

6 Norm-Sensitive Bridge Boundary

Figure 5 expands the projected-scalar bridge. The bridge detects scale perturbations that cosine-only projected-token sketches miss because it keeps norm information. However, a projection-aware null-space attack can avoid the one-dimensional projection. This is the boundary used in the main paper: norm-sensitive projections are useful hybrid components, not adaptive cryptographic guarantees.

7 Material-Tamper Coverage and Cosine Blind Spot

Figure 6 varies perturbation strength for the material-tamper study. Projcos4 detects Gaussian perturbations across the tested strengths, while scale-only perturbations remain undetected. This matches the TSTC mechanism: projected-token cosine sketches retain directional geometry but deliberately discard most norm information.

Table 6 gives the exact A/B and B/D attack-family TPRs summarized by the main paper’s material-tamper figure.

Table 7 summarizes the output-affecting subset across six stack pairs. It separates attacks that merely perturb checkpoint statistics from attacks that change the next-token neighborhood. Directional attacks remain detected by projcos4 on the output-affecting subset, while scale perturbations remain the cosine-only blind spot even when they affect the output neighborhood.

8 Checkpoint-Sketch Payload and Challenge-Time Runtime

Figure 7 visualizes the payload and latency numbers summarized in the main paper’s verifier-overhead table. Stronger checkpoint sketches increase reveal payload, but challenge-time verifier latency remains in the few-millisecond range for the measured implementation.

9 Full Queued-Workload Policy Metrics

Figure 8 returns to the queued-workload placement replay with the full metric set. Network-aware placement keeps scalar16 on heterogeneous paths, while risk-constrained and adaptive-verifier placement upgrade risky profile keys and remove infeasible usage without changing measured goodput.

Table 1. Measured verifier profiles consumed by the closed-loop placement replay. Each row is a stack-pair/sketch profile key over the three checked C1–C3 boundaries. B/tr. is the sketch payload per three-checkpoint challenge trace; TPR_p is primary material-tamper detection; TPR_s is scale-perturbation detection; and ms is honest challenge-time verifier latency per trace. “inadjud.” denotes an inadjudicable profile class; task feasibility is determined by the explicit admission test.

Pair	Sketch	B/tr.	FPR	TPR_p	TPR_s	LocAcc	ms	Risk
A/B	projcos16	3072	0.120	1.000	0.000	1.000	4.04	medium
A/B	projcos4	768	0.020	1.000	0.000	1.000	3.80	low
A/B	projcos8	1536	0.080	1.000	0.000	1.000	3.92	low
A/B	scalar16	192	0.150	0.760	0.520	0.760	3.60	high
A/B	scalar64	768	0.415	0.855	0.880	0.855	3.47	inadjud.
A/B-RTXint8	projcos16	3072	0.080	1.000	0.000	1.000	4.04	low
A/B-RTXint8	projcos4	768	0.020	1.000	0.000	1.000	3.80	low
A/B-RTXint8	projcos8	1536	0.035	1.000	0.000	1.000	3.92	low
A/B-RTXint8	scalar16	192	0.155	0.600	0.370	0.600	3.60	high
A/B-RTXint8	scalar64	768	0.520	0.855	0.880	0.855	3.47	inadjud.
A/C	projcos16	3072	0.135	1.000	0.000	1.000	4.04	medium
A/C	projcos4	768	0.045	1.000	0.000	1.000	3.80	low
A/C	projcos8	1536	0.075	1.000	0.000	1.000	3.92	low
A/C	scalar16	192	0.110	0.585	0.355	0.585	3.60	high
A/C	scalar64	768	0.305	0.875	0.930	0.875	3.47	inadjud.
A/D	projcos16	3072	0.065	1.000	0.000	1.000	4.04	low
A/D	projcos4	768	0.020	1.000	0.000	1.000	3.80	low
A/D	projcos8	1536	0.065	1.000	0.000	1.000	3.92	low
A/D	scalar16	192	0.090	0.510	0.220	0.510	3.60	high
A/D	scalar64	768	0.380	0.795	0.815	0.795	3.47	inadjud.
B/D	projcos16	3072	0.065	1.000	0.000	1.000	4.04	low
B/D	projcos4	768	0.075	1.000	0.000	1.000	3.80	low
B/D	projcos8	1536	0.075	1.000	0.000	1.000	3.92	low
B/D	scalar16	192	0.245	0.950	0.720	0.950	3.60	high
B/D	scalar64	768	0.590	1.000	0.995	1.000	3.47	inadjud.
E/F	projcos16	3072	0.030	1.000	0.005	1.000	4.04	low
E/F	projcos4	768	0.105	1.000	0.060	0.985	3.80	medium
E/F	projcos8	1536	0.005	1.000	0.000	1.000	3.92	low
E/F	scalar16	192	0.165	1.000	1.000	1.000	3.60	medium
E/F	scalar64	768	0.615	1.000	1.000	1.000	3.47	inadjud.
F/F	scalar16	192	0.000	1.000	1.000	1.000	3.60	low
F/F	projcos4	768	0.000	1.000	1.000	1.000	3.80	low
F/F	projcos8	1536	0.000	1.000	1.000	1.000	3.92	low
F/F	projcos16	3072	0.000	1.000	1.000	1.000	4.04	low

Table 2. Fixed inputs for the controlled placement replay used in the main evaluation.

Item	Setting
Candidate placements	13 total: 1 homogeneous F/F and 12 heterogeneous LAN/WAN variants
Stack-pair labels	A/B, A/B-RTXint8, A/C, A/D, B/D, E/F, and F/F control
Verifier profiles	34 measured profile keys, each identified by stack pair and sketch mode over C1–C3
Workload prompts	200 held-out evaluation prompts
Arrival patterns	single: one task every 5.0 s; queued-8: eight tasks every 0.25 s
Replay seeds	5 seeds: 20260513–20260517
Target sweep	$\alpha \in \{0.05, 0.10, 0.20\}$, $\beta = 0.90$
Main target	$\alpha = 0.10$, $\beta = 0.90$
Sketch budget	$S^{\max} = 3072$ bytes per three-checkpoint trace
Verifier budget	$C^{\max} = 5.0$ ms per challenge
Challenge probability	0.10, used for false-risk and challenge-cost expectations
Group-size bound	$1 \leq \mathcal{G} \leq 4$; heterogeneous candidates use group size 3

Table 3. Queued-8 closed-loop placement replay at $\alpha = 0.10$, $\beta = 0.90$. Risk-constrained hard admission and adaptive verifier-mode selection remove infeasible profile keys before disclosure; queue-aware scoring improves modeled latency and goodput, but does not by itself enforce verifiability. Chal. ms/task is expected challenge-time verifier cost under challenge probability 0.10, and FD risk is expected false-dispute risk.

Replay	Policy	Med. task s	P95 task s	Goodput	Chal. ms/task	FD risk	Infeas.
controlled	network-aware, projcos4	90.156	170.384	1.079	.380	.0105	1.000
controlled	risk-constrained, projcos4	91.844	173.576	1.060	.380	.0020	.000
controlled	network-aware, adaptive	90.156	170.384	1.079	.392	.0005	.000
controlled	homogeneous only	119.301	225.498	.822	.360	.0000	.000
queue-aware	network	5.522	9.757	12.266	.360	.0142	.935
queue-aware	risk-constrained	7.004	12.714	10.392	.380	.0033	.000
queue-aware	adaptive	5.457	9.781	12.251	.380	.0028	.000



Figure 1. Selected verifier-profile risk composition in the placement replay. Network and queue-aware policies can select infeasible profile keys; the risk-constrained and adaptive policies filter or upgrade those choices before task disclosure.

Table 4. Selective-delivery group-size sweep for a 100MB encrypted payload. Latency is all-providers-ready time under the calibrated LAN/WAN transfer model; requester egress counts payload-sized transfers sent by the requester.

Network	k	RPD med. s	PPD med. s	Med. red.	RPD req. MB	PPD req. MB
LAN	1	0.864	1.719	−99.1%	100.0	100.001
LAN	2	1.715	1.720	−0.3%	200.0	100.001
LAN	4	3.417	1.719	49.7%	400.0	100.002
LAN	8	6.820	1.719	74.8%	800.0	100.005
WAN	1	20.088	30.259	−50.6%	100.0	100.001
WAN	2	40.099	30.272	24.5%	200.0	100.001
WAN	4	80.088	30.267	62.2%	400.0	100.002
WAN	8	160.100	30.267	81.1%	800.0	100.005

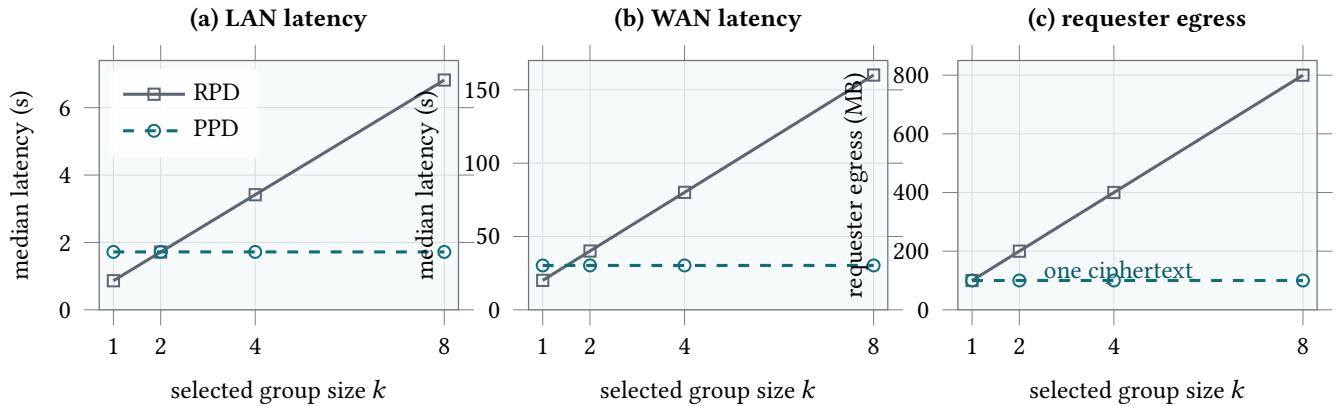


Figure 2. Placement-defined selective-delivery scaling. RPD becomes dominated by repeated payload-sized requester transfers as k grows; PPD keeps requester egress close to one ciphertext plus small access packages.

Table 5. Live local-store validation for a 100MB payload. Latency is local-store specific; the key result is fresh-object semantics and requester-egress scaling. PPD store MB is aggregate off-chain-store egress.

k	RPD med. s	PPD med. s	RPD req. MB	PPD req. MB	PPD store MB
1	0.993	1.033	100.0	100.001	100.0
2	0.997	1.034	200.0	100.001	200.0
4	1.048	1.046	400.0	100.003	400.0
8	1.065	1.089	800.0	100.005	800.0

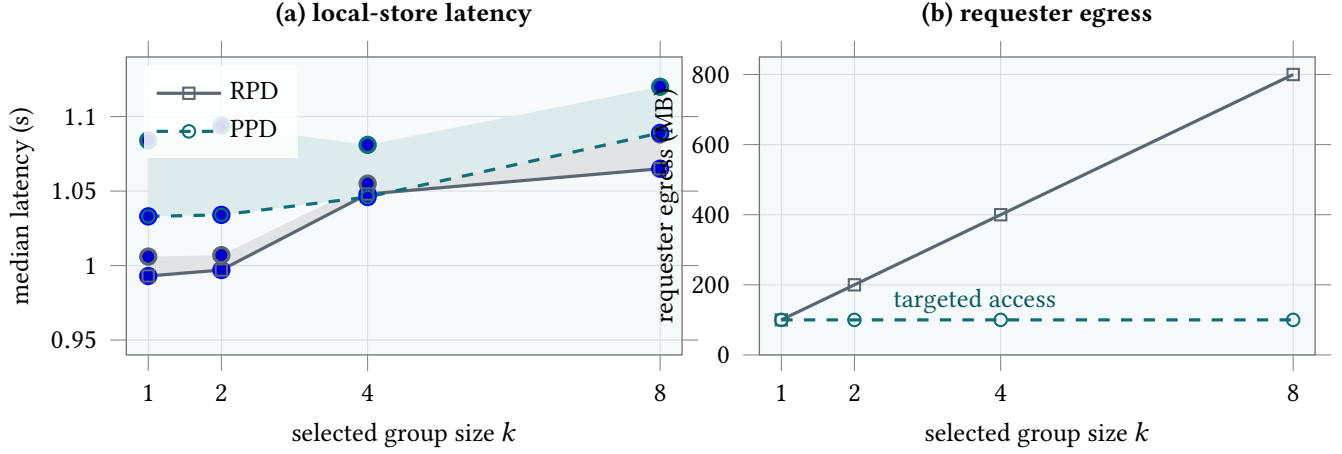


Figure 3. Live local-store validation of selective delivery. The plot confirms fresh content-hash object storage and the requester-egress separation between RPD and PPD; its latency is local-store validation evidence rather than a WAN/LAN deployment result.

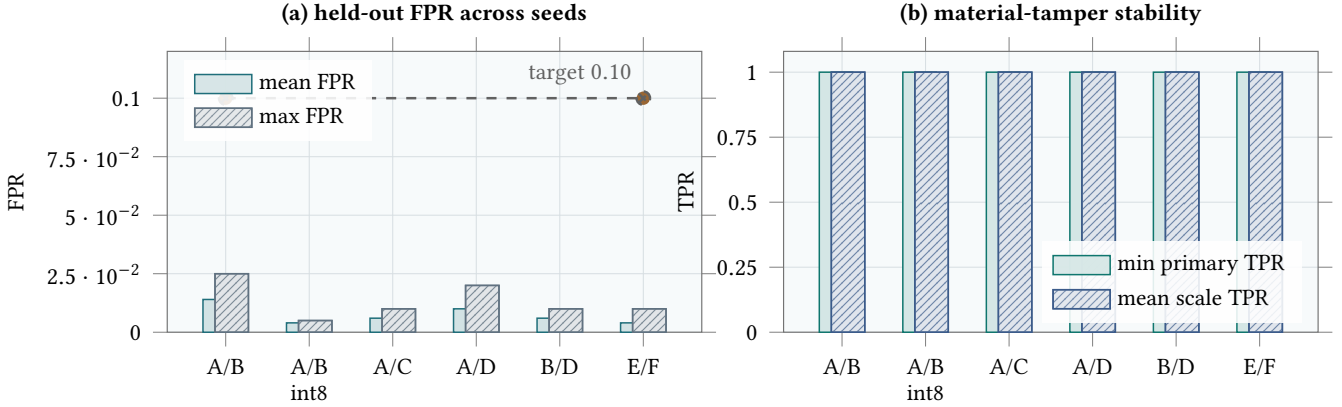


Figure 4. Projected-scalar multiseed stability. The plot summarizes A/B, A/B-RTXint8, A/C, A/D, B/D, and E/F across five seeds, showing that projscalar1_abs remains low-FPR while preserving primary material-tamper detection.

Table 6. Material-tamper TPR by attack family. Projected-token cosine sketches detect direction-changing attacks but are weak against direction-preserving scale perturbations.

Pair	Variant	Gauss.	Cross prompt	Wrong shard	Layer skip	Scale
A/B	scalar16	1.00	.76	.76	.83	.52
A/B	projcos4	1.00	1.00	1.00	1.00	.00
B/D	scalar16	1.00	.95	.95	1.00	.72
B/D	projcos4	1.00	1.00	1.00	1.00	.00

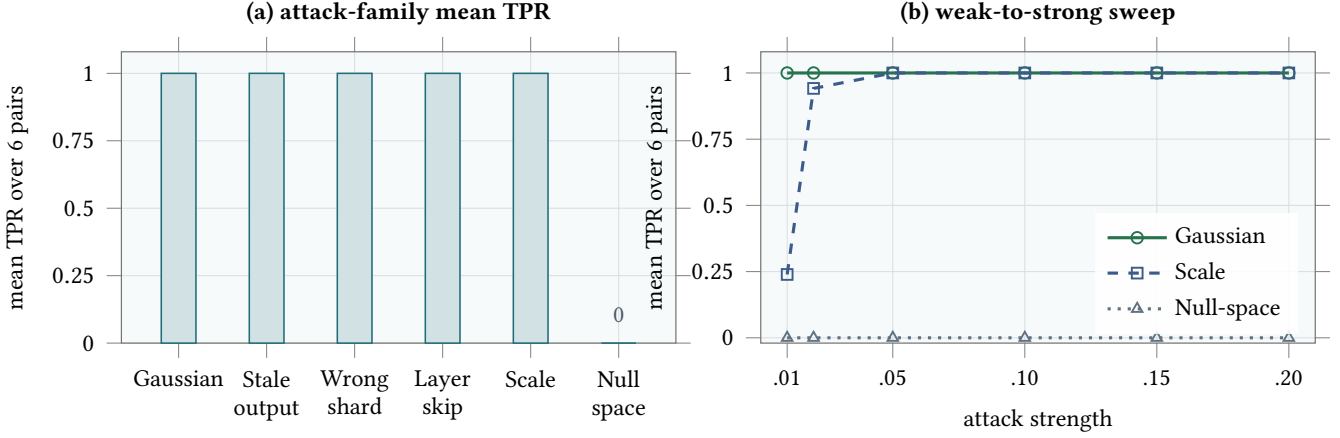


Figure 5. Projected-scalar attack-family and strength-sweep results. The sketch covers scale perturbations that cosine gaps miss, but exposes the expected null-space boundary for projection-aware attacks.

Table 7. Output-affecting material-tamper subset across six stack pairs. OA is the fraction of prompts whose next-token top-5 neighborhood changes under the attack; TPR ranges are computed only over this output-affecting subset.

Attack family	OA rate	Projcos4 OA TPR	Scalar16 OA TPR	Mean top-5 Jaccard
Gaussian	1.000	1.000–1.000	1.000–1.000	0.002
Cross-prompt stale	0.941	1.000–1.000	0.558–1.000	0.502
Wrong-shard output	0.941	1.000–1.000	0.558–1.000	0.502
Layer skip	1.000	1.000–1.000	0.510–1.000	0.173
Scale perturbation	0.389	0.000–0.064	0.228–1.000	0.893

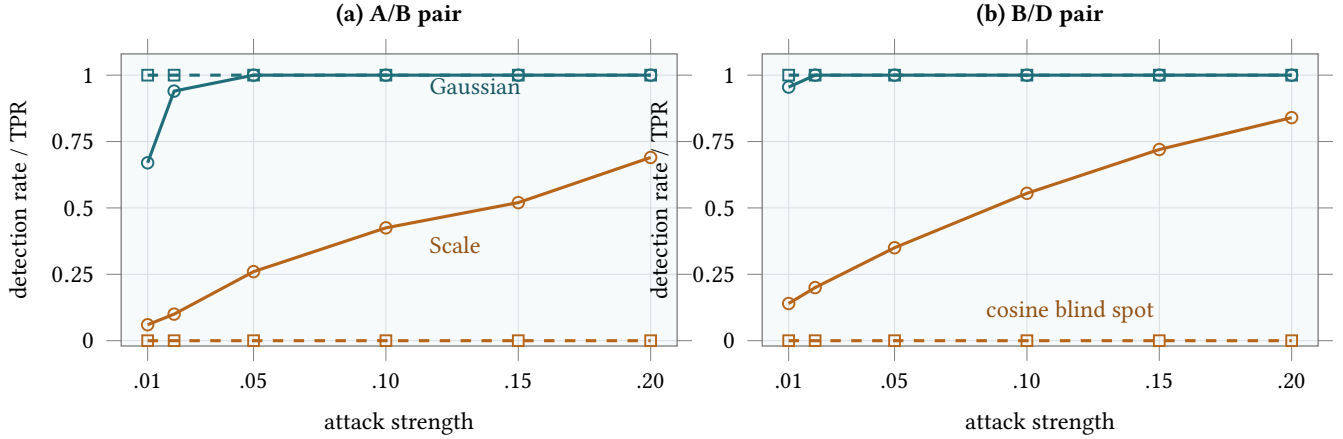


Figure 6. Material-tamper strength sweep. Solid lines are scalar16 and dashed lines are projcos4; teal denotes Gaussian noise and orange denotes scale perturbations. The sweep confirms that cosine-only projected-token sketches detect direction-changing perturbations but remain weak against direction-preserving scale changes.

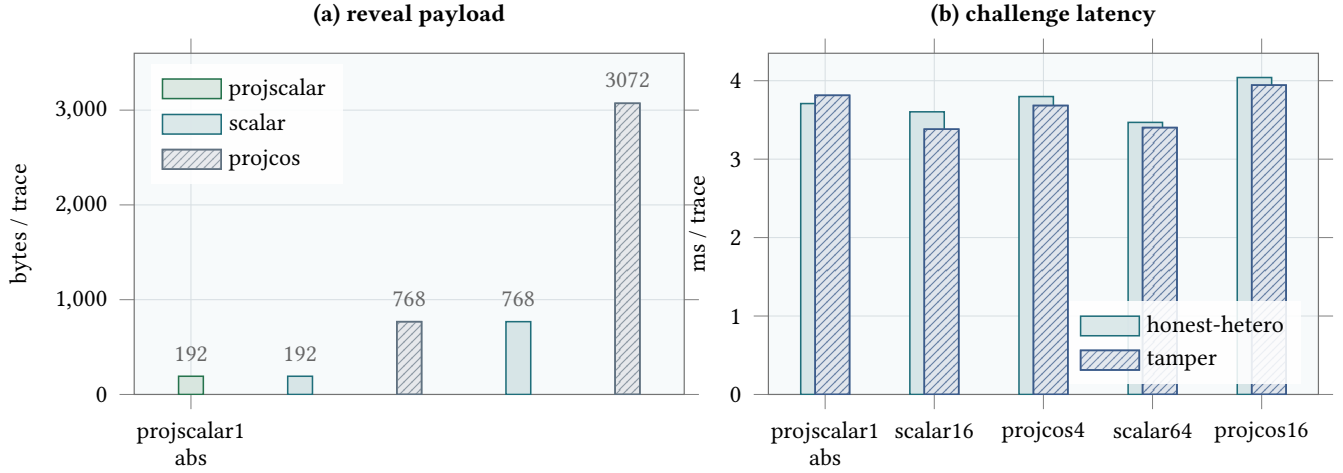


Figure 7. Checkpoint-sketch payload and challenge-time verifier latency. Reveal payload scales with sketch dimensionality, while measured verifier latency remains in the few-millisecond range across scalar-coordinate, projected-token, and projected-scalar variants.

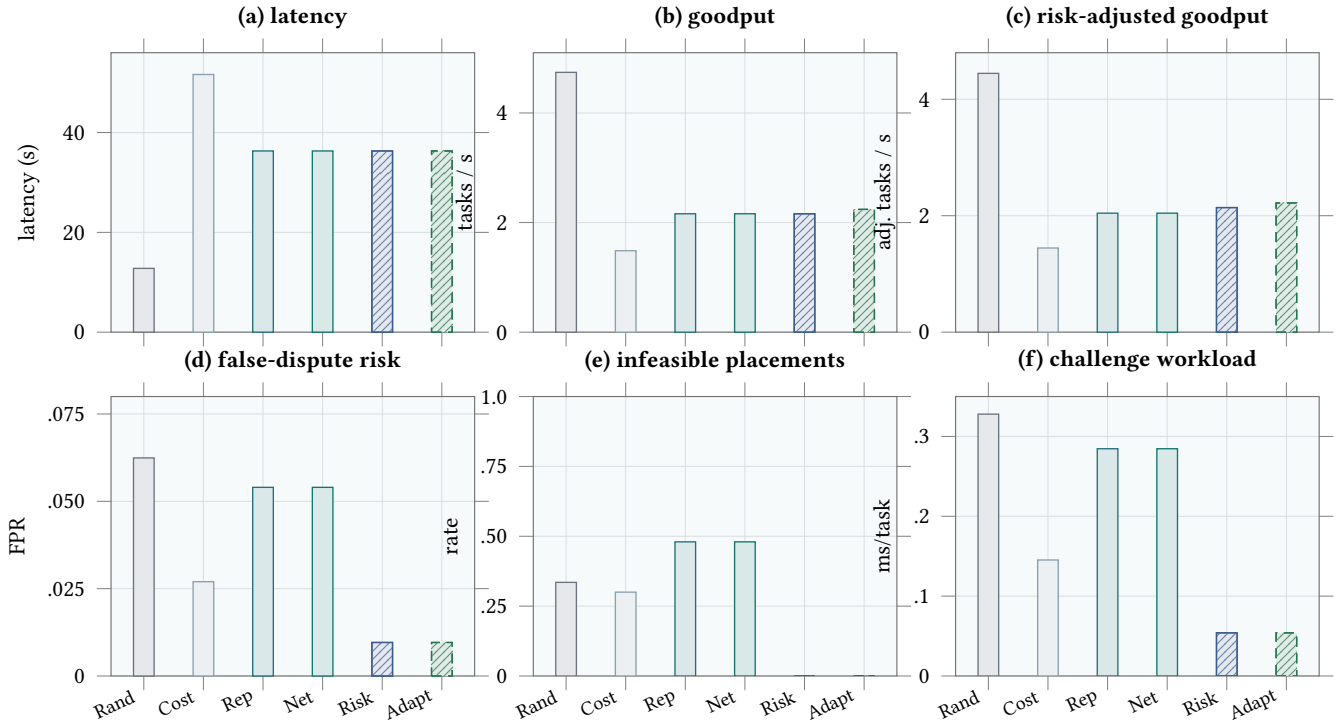


Figure 8. Queued-workload policy comparison. The plot shows the tradeoff among modeled latency, goodput, risk-adjusted goodput, false-dispute risk, infeasible usage, and expected challenge workload; Risk and Adapt. denote the risk-constrained and adaptive-verifier policies.